Sahaj Bhadja

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Summary

Application developer pursuing an MSCS at USC with 2+ years building **SQL-backed applications**, data-processing workflows, and internal tools, with hands-on experience in **T-SQL-adjacent querying**, production troubleshooting, reporting, and healthcare-adjacent systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **SQL-backed platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning query patterns, and improving responsiveness for analytics workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving reliability, release confidence, and troubleshooting visibility across recurring production data-processing pipelines.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, accurate data movement, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating stakeholder requirements into maintainable backend services while improving supportability, implementation ownership, and production debugging workflows.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations through concurrent execution, CI/CD-backed releases, and cost-aware debugging.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask booking and payment APIs** for a cross-platform services product, supporting 50K+ booking records while maintaining sub-200ms core flows and reliable state synchronization across user sessions.
- Integrated **payment, coupon, and webhook workflows** with Firebase-backed synchronization, reducing manual handling by **70%** while improving issue traceability and operational reliability for external customer experiences.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening responsiveness for real-time user-facing applications.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: SQL, Java, Python, JavaScript/TypeScript, C++, Bash
Databases & BI: MS SQL/SQL Server, PostgreSQL, MySQL, MongoDB, schema design, query optimization, reporting workflows
Backend & Integration: REST APIs, Flask, FastAPI, Node.js, ETL pipelines, data exchange workflows
Applications & Support: production troubleshooting, code/process documentation, technical specifications, change-oriented delivery
Cloud & Delivery: AWS, Azure, Docker, Jenkins, Git, CI/CD
Engineering: unit testing, integration testing, Agile/Scrum, cross-functional collaboration, data analysis

Projects

HMIS Live | *MERN, TypeScript, Twilio, Appwrite*
- Built a **healthcare management platform** with appointment workflows, notifications, and dashboard experiences, reducing administrative workload by **30%** through secure automation and structured system behavior.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring and schema exploration through a unified **Java** and **TypeScript** codebase with Docker-backed local workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.